

STAGE – 4

PYTHON DOCUMENTATION & INSIGHT PRESENTATION

Project Title

“Analysing Growth Patterns of Limited Liability Partnerships Across States”

Objective

The objective of this project is to analyse LLP registration data to identify registration trends, compare state-wise business activity, understand LLP status distribution, and generate business insights that support data-driven decision-making.

Dataset Information

The project uses the LLP Registration dataset containing information such as registration dates, state names, LLP status, number of partners, and contribution amounts. The data was collected from a reliable government source and used for business analysis.

Tools & Technologies

- Python
- Google Collab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI (Dashboard)

Data Cleaning & Preprocessing

The dataset was cleaned by removing duplicate records, handling missing values, converting date columns into datetime format, and creating additional features such as registration year for analysis.

Exploratory Data Analysis

EDA was performed to understand LLP registration trends, state-wise distribution, LLP status, partner information, and contribution amounts using descriptive statistics and charts.

Visualizations Used

The project includes:

STAGE – 4

PYTHON DOCUMENTATION & INSIGHT PRESENTATION

- Bar Chart (State-wise Registrations)
- Line Chart (Year-wise Trend)
- Pie Chart (LLP Status Distribution)
- Histogram (Partner Distribution)
- Box Plot (Outlier Detection)
- Correlation Heatmap
- Pair Plot
- Pivot Table
- Power BI Dashboard

Challenges Faced

1. Handling missing and inconsistent data.
2. Converting date fields into the correct format.
3. Removing duplicate records.
4. Managing outliers in numerical data.
5. Creating clear and meaningful visualizations.
6. Interpreting results from multiple charts and presenting actionable business insights.

Conclusion

The project successfully analysed LLP registration data using Python and EDA techniques. It identified important business trends, regional registration patterns, and LLP characteristics through data visualization. The insights generated can support policymakers, investors, entrepreneurs, and business analysts in making informed decisions. The project demonstrates how data analytics transforms raw data into actionable business intelligence.